

Mohini Sharma

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PROFESSIONAL SUMMARY

ML/AI engineer specializing in computer vision and LLMs. Built defect detection, 3D reconstruction, and documentation platforms handling 10K+ images in production. AWS re:Invent showcase. Track record: 60%+ workflow reduction, 5x system scaling. Cross-domain innovator: hardware-software integration, GenAI pipelines, multi-agent automation.

EDUCATION

B.Tech, Computer Science Engineering • CGPA: 8.46

Acropolis Institute of Technology and Research, Indore

Intermediate (XII) 2021 • 90.6%

Kalyan Mateshwari Public School, Indore

SKILLS

Languages: Python, C++, Dart, JavaScript

Frameworks & Tools: PyTorch, YOLOv8, Detectron2, LangChain, LangGraph, OpenCV, Flask, FastAPI, React, Flutter, Streamlit, MLflow, Docker, AWS SageMaker, Power BI, Firebase, Azure

AI Domains: Computer Vision, NLP, Generative AI, Agentic AI, 3D Reconstruction, RAG Pipelines, MLOps, Intelligent Document Processing

EXPERIENCE

Software Developer — AI Research & Development | Ascentt AI

Feb 2025 – Present, Indore

- **Computer Vision & Defect Detection** — Built vehicle defect assessment system (10K+ images, 94% precision). Innovation: LiDAR-depth fusion for quantitative severity scoring. Result: Reduced inspection time 4hrs → 45mins; scaled 500 → 2,500 units/day.
- **Generative AI & 3D Reconstruction** — Architected Aura Nova pipeline for photorealistic 3D vehicle reconstruction (TRELIS, InstantMesh). Enables AR defect visualization. AWS re:Invent global showcase selection.
- **Agentic AI & LLM Automation** — Multi-agent LangGraph workflows for repair guidance + doc processing. ASC-Cadence LLM-CAD/FEA pipeline: reduced design cycle 40%. Power BI HR chatbot (Ollama): eliminated external LLM costs + privacy risks.
- **MLOps & Systems** — AutoML pipeline (EDA → training → evaluation). Deployed on AWS SageMaker + MLflow. Reduced training feedback loop 3 days → 4 hours via automated experiment tracking.

Data Science Intern | Uoons E-commerce Pvt Ltd.

May 2024 – Jul 2024, Indore

- Built Power BI dashboards reducing report generation 2hrs → 20mins. KNN recommendation model (76% accuracy); drove 20% engagement uplift.

PROJECTS

Anuvaad — AI Multilingual Media Translation App •

play.google.com/store/apps/details?id=com.mohinisharma.t_to_v

- 100+ downloads; 4.2★ rating. Converts press releases to multilingual videos (13 Indian languages + English) cutting manual effort by 60%.
- Innovation: Article-to-video generation + auto language detection enables unmet use case for Indian media distribution.
- End-to-end execution: backend (Flask NLP/OCR/ASR-TTS) → mobile (Flutter) → deployment (Azure, Firebase). Icreate research grant (April 2024).

DiagramStudio — AI Technical Documentation Platform • <https://diagramstudio.in/>

- AI generator: 6 diagram types (flowchart, architecture, ERD, swimlane, high-level, sequence) + 9 doc types (TDD, PRD, API ref, RFC, ADR, runbook, onboarding, post-mortem, general).
- Bidirectional diagram-as-code: edit canvas or DSL text; changes sync instantly. Backed by Gemini LLM + heuristic guardrails. Diff-friendly + version-control ready.
- GitHub integration: import repos → auto-generate architecture diagrams + docs from actual code. 3D repo-graph visualization; live multiplayer collab; role-based access.
- Multi-format export (PNG, SVG, Draw.io, PDF, DOCX). Structural Notes system for diagram elements. Free/Pro/Team pricing with Razorpay.
- Tech: React/Vite + hand-built ELK layout engine, FastAPI + Gemini API, DynamoDB, Firebase Auth, GitHub Actions → AWS ECR/EC2.

Dark Pattern Buster Chrome Extension • github.com/mohinisharma4410/DPBH_2023_imp

- ML-powered Chrome extension detecting dark patterns on e-commerce platforms; shortlisted Top 10 at DPBH 2023 (IIT-BHU).
- Captured 11 dark pattern types across e-commerce websites + insurance policies. Raises awareness of deceptive UX design. Tech: Python, ML, Flask, JavaScript, BeautifulSoup.

ACHIEVEMENTS

- AWS re:Invent global showcase (2024): Aura Nova 3D reconstruction + Overbody Damage Detection POCs selected for industry visibility
- Top 10 — Dark Pattern Buster Hackathon 2023 (IIT-BHU) | Top 14 — IIT Drishti SHAKTI 2.0 (IIT Indore)
- Icreate research grant (April 2024): Anuvaad multilingual media translation project